Define LLM from openrputer

In [124]:
```
from langchain_openai import ChatOpenAI

llm = ChatOpenAI(
    api_key="sk-or-v1-
891312635c0db875a91dea9419f2c20a0177a90807a2ca2261f035bd5eea6c4f",
    base_url="https://openrouter.ai/api/v1",
    model="openrouter/free",
    temperature=0.8
)
```

Prompt Templates in LangChain

In [ ]:
```
# step 1 : import package
from langchain_classic.prompts import PromptTemplate

# define our template

text = """hello my name is {name}
can you explain topic about {topic}
and you should answer in {num_of_words} words only , and in
{language}"""

template = PromptTemplate.from_template(text)


# define prompts

prompt_1 = template.format(name= "sayed",topic ="microwave",
num_of_words="100",language="arabic")
print(prompt_1)
```

```
hello my name is sayed
can you explain topic about microwave
and you should answer in 300 words only , and in arabic
```

In [143]:
```
response = llm.invoke(prompt_1)
```

In [145]:
```
from IPython.display import display , Markdown

display(Markdown(response.content))
```

```
<IPython.core.display.Markdown object>
```

System Prompting

In [147]:
```
from langchain_classic.schema import SystemMessage , HumanMessage

system_prompt = SystemMessage("you are alien , your answers should be in
arabic only even if the Question in English, and you should answer with
10 words only, if user asks you any Question not related with math you
should say ( I don't know )")

prompt = HumanMessage("what is the meaning of life ?")

total  = [system_prompt,prompt]

response = llm.invoke(total).content

print(response)
```

```
لا أعرف
```

Few Shot Prompting

In [150]:
```
# part 1 : define llm
from langchain_openai import ChatOpenAI
llm = ChatOpenAI(
    api_key="sk-or-v1-
c16747edfd6ad7e20f561e1a3920483b0eca7725abe9324025edfbbfc22e21aa",
    base_url="https://openrouter.ai/api/v1",
    model="openrouter/free",
    temperature=0.8
)

# part 2 : import packages
from langchain_classic.prompts import PromptTemplate ,
FewShotPromptTemplate

# part 3 : define prompt template
temp = PromptTemplate.from_template("the [{word}] and output is
[{function}]")

examples = [
    {"word" : "cat" , "function" : "t-a-c"},
    {"word" : "sky" , "function" : "y-k-s"},
    {"word" : "lion" , "function" : "n-o-i-l"},
    {"word" : "microwave" , "function" : "e-v-a-w-o-r-c-i-m"},
    {"word" : "omar" , "function" : "r-a-m-o"}
]

fewshot = FewShotPromptTemplate(
    example_prompt = temp,
    examples = examples,
    suffix = "now for [{word}]",
    input_variables = ["word"]
)

# part 4 : create prompt
prompt = fewshot.format(word="school is clear")

request = llm.invoke(prompt).content
print(request)
```

For “school is clear”

Reverse the entire text (including the spaces) and separate each character with a hyphen:

```
r-a-e-l-c- -s-i- -l-o-o-h-c-s
```

So the output string is: **[r-a-e-l-c- -s-i- -l-o-o-h-c-s]**.

In [151]:
```
from IPython.display import display , Markdown

display(Markdown(request))
```

```
<IPython.core.display.Markdown object>
```

Comma Separated List Output Parser

In [158]:
```
instruction = "Your response should be a list of comma separated values,
eg: `foo, bar, baz` or `foo,bar,baz`"

temp = "state 10 countries in Africa \n{instruction}"
template = PromptTemplate.from_template(temp)

prompt = template.format(instruction = instruction)

request = llm.invoke(prompt).content
print(request)
```

```
Egypt,South Africa, Nigeria, Kenya, Morocco, Ghana, Algeria,
Tanzania, Uganda, Ethiopia
```

In [160]:
```
out = request.strip()
out = out.split(", ")
print(out)
```

```
['Egypt,South Africa', 'Nigeria', 'Kenya', 'Morocco', 'Ghana',
'Algeria', 'Tanzania', 'Uganda', 'Ethiopia']
```

Memory in LangChain

In [164]:
```
# step 1 : define model
from langchain_openai import ChatOpenAI
llm = ChatOpenAI(
    api_key="sk-or-v1-
c16747edfd6ad7e20f561e1a3920483b0eca7725abe9324025edfbbfc22e21aa",
    base_url="https://openrouter.ai/api/v1",
    model="openrouter/free",
    temperature=0.8
)

# step 2 : define converstion
from langchain_classic.chains import ConversationChain
from langchain_classic.memory import ConversationBufferMemory

chat = ConversationChain(
    llm = llm,
    memory = ConversationBufferMemory()
)

response = chat.predict(input="who Iam ?")
```

In [165]:
```
print(response)
```

```
AI: Hmm, that's an interesting question! Since I don’t have access
to personal information about you unless you share it with me, I’m
not sure who you’re referring to. Could you clarify what you mean
by "who I am"? Are you asking about your own identity, or perhaps
someone else? I’d love to help, but I need a bit more context to
give you a meaningful answer! 😊
```

In [166]:
```
response = chat.predict(input="my name is omar ahmed and I'm 25 years
old")
```

In [167]:
```
print(response)
```

```
Hello, Omar Ahmed! Nice to meet you. I’ve noted that you’re 25
years old. How can I assist you today? Whether you have questions,
need help with something, or just want to chat, I’m here for you!
Let me know how I can be of service. 😊
```

In [168]:
```
response = chat.predict(input="what do you know about me ?")
```

In [169]:
```
print(response)
```

That's a great question, Omar! Based **only** on what you've shared in our conversation so far, I know the following about you:

1. **Your Name:** Omar Ahmed
2. **Your Age:** 25 years old

That's it! 😊 I don't have access to any other personal information about you unless you choose to share it in this chat. I don't store data from our conversations beyond the current session unless you explicitly ask me to remember something for later (and even then, it's limited to what *you* tell me to remember).

So, the full picture I have right now is that you're a 25-year-old named Omar Ahmed chatting with me. It's nice to get to know you a little better!

Is there anything specific you'd like me to remember for our next chat, or is there something else you'd like to tell me about yourself? I'm here to help! 😄

RAG

In [ ]:
```
## step 1 : create vector database
        # Loading
        # Chunking
        # Embedding
        # Vector stores

# Loading
from langchain_classic.document_loaders import TextLoader
loader = TextLoader("text.txt")
document = loader.load()

# Chunking
from langchain_classic.text_splitter import TokenTextSplitter

splitter = TokenTextSplitter(
        chunk_size = 200,
        chunk_overlap = 20
)

chunks = splitter.split_documents(document)


# Embedding
from langchain_community.embeddings import HuggingFaceEmbeddings

embeddings = HuggingFaceEmbeddings(
        model_name = "sentence-transformers/all-MiniLM-L6-v2",
        model_kwargs = {"device":"cpu"}
)

vectors = embeddings.embed_documents([i.page_content for i in chunks])

# vector store
from langchain_classic.vectorstores import FAISS

vectorDB = FAISS.from_documents(chunks,embeddings)
```

```
Loading weights: 100%|████████| 103/103 [00:00<00:00,
2024.99it/s]
```

In [114]:
```
query = "what are programs in Faculty of Engineering Alexandira University"

similar_docs = vectorDB.similarity_search_with_score(query,k=3)

for i in similar_docs :
    print(i[0].page_content)
```

with opportunities to experiment, simulate systems, and develop prototypes for modern engineering applications.

Student life inside the faculty is highly active and competitive. Many students participate in technical competitions, robotics teams, programming contests, and student organizations. Activities such as scientific conferences, engineering workshops, and industrial training programs help students connect academic learning with professional experience. These activities also improve leadership, communication, and project management skills, which are essential for successful engineering careers.

Graduates of the faculty work in many sectors including telecommunications, construction, software development, energy systems, embedded systems, manufacturing, and academic research. Some graduates continue their studies in international universities, while others contribute directly to industry and entrepreneurship. The faculty has built a reputation for producing engineers with strong analytical abilities, technical competence, and adaptability to modern technological changes.

Overall, the Faculty of Engineering Alexandria University represents an important center for engineering education and scientific advancement. It continues to contribute to technological innovation, industrial growth, and human development by preparing
The Faculty of Engineering Alexandria University is considered one of the oldest and most respected engineering institutions in Egypt and the Middle East. Established as part of Alexandria University, the faculty has played a major role in preparing generations of engineers, researchers, and innovators who contributed to industrial, technological, and scientific development across many fields. Located in the historic city of Alexandria, the faculty combines a long academic tradition with modern engineering education and research.

The faculty offers a wide range of engineering programs that cover both classical and advanced engineering disciplines. Students can specialize in departments such as Electrical Engineering, Mechanical Engineering, Civil Engineering, Architecture, Chemical Engineering, Industrial Engineering, Computer Engineering, and Electronics and Communications Engineering. Each department focuses on building strong theoretical foundations while also emphasizing practical applications, laboratory work, technical projects, and problem-solving skills. This balance between theory and practice helps students understand how engineering concepts are applied in real-world systems and industries.

One of the most important characteristics of the faculty is
 in real-world systems and industries.

One of the most important characteristics of the faculty is its strong academic environment. Students are encouraged to think critically, analyze complex systems, and develop innovative solutions for engineering challenges. Engineering education at the faculty is known for being rigorous and demanding, requiring students to master mathematics, physics, programming, electronics, mechanics, and system design. Through assignments, workshops, and graduation projects, students gain experience in teamwork, research methodologies, technical writing, and engineering analysis.

Research is also an essential part of the faculty’s identity. Professors and researchers work on projects related to renewable energy, telecommunications, artificial intelligence, robotics, power systems, structural engineering, biomedical engineering, and advanced manufacturing technologies. Many research activities aim to solve industrial and societal problems while supporting technological progress in Egypt and beyond. Laboratories and research centers within the faculty provide students and researchers with opportunities to experiment, simulate systems, and develop prototypes for modern engineering applications.

Student life

In [170]:

```
from langchain_core.prompts import PromptTemplate

temp = """
you are assistant, Answer the next Question using provided context,
If you don't know the answer, just say you don't know.
answer should be within 200 words or lower only

## context :
{context}

## Question :
{Question}
"""

temp = PromptTemplate.from_template(temp)


query = "what are programs in Faculty of Engineering Alexandira
University"

similar_docs = vectorDB.similarity_search_with_score(query,k=4)

context = []
for i in similar_docs :
    context.append(i[0].page_content)


prompt = temp.format(context="\n".join(context), Question = query)
print(prompt)
```

```
you are assistant, Answer the next Question using provided
context,
If you don't know the answer, just say you don't know.
answer should be within 200 words or lower only

## context :
 with opportunities to experiment, simulate systems, and develop
prototypes for modern engineering applications.

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teams, programming contests, and student organizations. Activities
such as scientific conferences, engineering workshops, and
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engineers with strong analytical abilities, technical competence,
```

and adaptability to modern technological changes.

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Student life
 and scientific advancement. It continues to contribute to technological innovation, industrial growth, and human development

```
by preparing skilled engineers capable of solving complex
challenges in an increasingly advanced and interconnected world.


## Question :
what are programs in Faculty of Engineering Alexandira University
```

In [122]:
```
response = llm.invoke(prompt).content
```

In [123]:
```
from IPython.display import display , Markdown

display(Markdown(response))
```

```
<IPython.core.display.Markdown object>
```

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